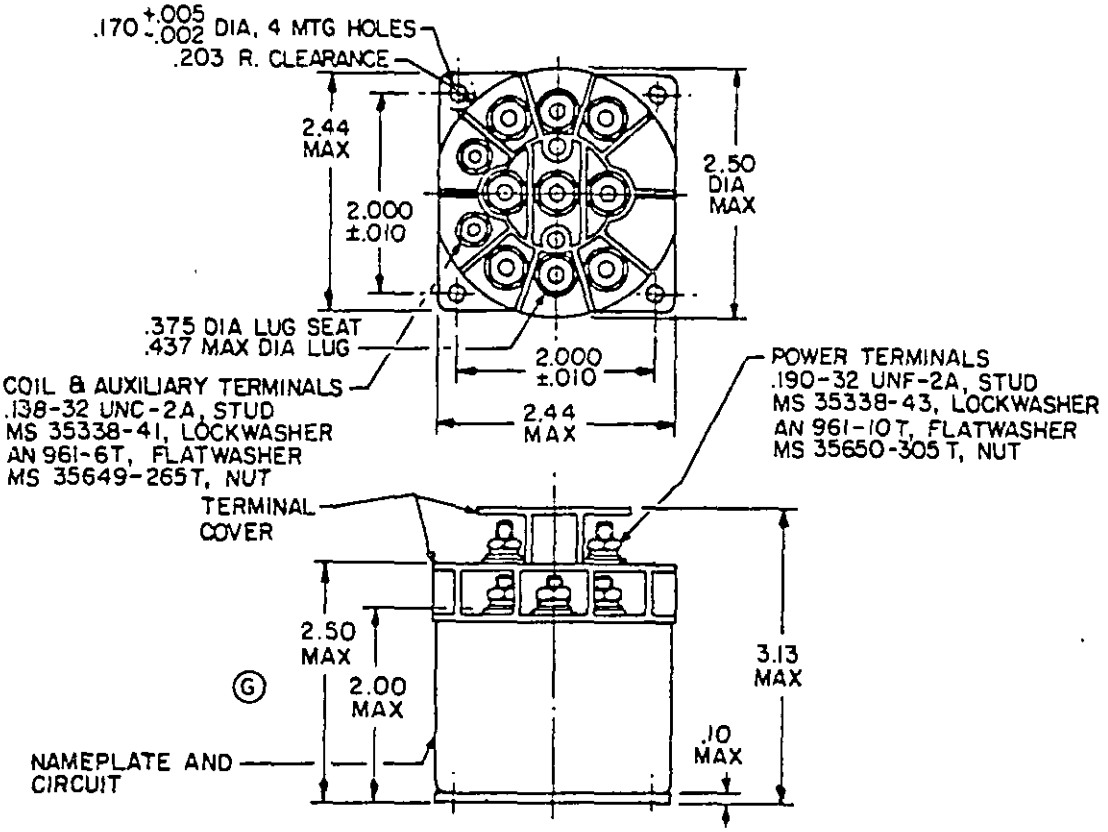
FED. SUP CLASS
5945



INCHES	MM	INCHES	MM
.002	0.05	.375	9.52
.005	0.13	.437	11.10
.010	0.25	1.91	48.5
.031	0.79	2.000	50.80
.10	2.5	2.44	62.0
.138	3.51	2.45	62.2
.170	4.32	2.50	63.5
.190	4.83	3.13	79.5
.203	5.16		

NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerance is ±.031 (0.79 mm).
3. Polarity indication applies to dc coils only.
4. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.
5. Referenced Government documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this standard to the extent specified herein.
6. Dc versions of this relay must not operate or be damaged by reverse polarity, semiconductors shall not be used for this purpose.
7. Permanent magnet drive consists of permanent magnet with its flux path switched and combined with the electro-magnetic flux.
8. Metric equivalents are given for general information only.

Ⓖ Denotes change

APPROVED 17 JAN 72 REVISED Ⓕ 22 APR 83 Ⓖ 29 APR 85

Review activity: AF-99

This military standard is approved for use within the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

P.A. AF-85 Other Cust	International interest	TITLE RELAYS, ELECTROMAGNETIC, 60 AMPERES, 3 PST OR 3 PDT, HERMETICALLY SEALED, PERMANENT MAGNET DRIVE Ⓖ	MILITARY STANDARD MS27751 (85)
Procurement Specification MIL-R-6106		SUPERSEDES:	PAGE 1 OF 7

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5945-F595

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TABLE I. Relay characteristics.

MS PART NUMBER	TYPE	COIL	CONTACTS		TERMINAL TYPE	MAX WEIGHT IN POUNDS 1/
			MAIN	AUX		
MS 27751-1	I	DC	3PDT	NONE	STUD	.85
MS 27751-2	I	AC	3PDT	NONE	STUD	.85
MS 27751-3	I	DC	3PNO	NONE	STUD	.85
MS 27751-4	I	AC	3PNO	NONE	STUD	.85
MS 27751-5	I	DC	3PNO	1NO-1NC	STUD	.85
MS 27751-6	I	AC	3PNO	1NO-1NC	STUD	.85
MS 27751-7	I	DC	3PDT	1NO	STUD	.85
MS 27751-8	I	AC	3PDT	1NO	STUD	.85
MS 27751-9	I	DC	3PDT	1NC	STUD	.85
MS 27751-10	I	AC	3PDT	1NC	STUD	.85
MS 27751-11	I	DC	3PDT	NONE	STUD	.85
MS 27751-12	I	AC	3PDT	NONE	STUD	.85
MS 27751-13	I	DC	3PNO	NONE	STUD	.85
MS 27751-14	I	AC	3PNO	NONE	STUD	.85
MS 27751-15	I	DC	3PNO	1NO-1NC	STUD	.85
MS 27751-16	I	AC	3PNO	1NO-1NC	STUD	.85
MS 27751-17	I	DC	3PDT	1NO	STUD	.85
MS 27751-18	I	AC	3PDT	1NO	STUD	.85
MS 27751-19	I	DC	3PDT	1NC	STUD	.85
MS 27751-20	I	AC	3PDT	1NC	STUD	.85
MS 27751-21	I	DC	3PDT	NONE	STUD	.85

1/ Weight includes terminal barriers.

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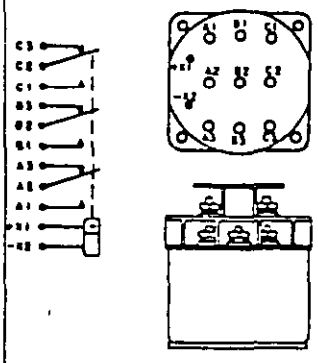
P A AF-85 Other Cust	International interest	TITLE RELAYS, ELECTROMAGNETIC, 60 AMPERES, 3 PST OR 3 PDT, HERMETICALLY SEALED, PERMANENT MAGNET DRIVE	MILITARY STANDARD MS27751 (85)
Procurement Specification MIL-R-6106		SUPERSEDES	PAGE 2 OF 7

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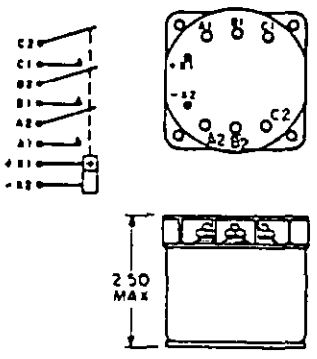
FED. SUP CLASS
5945

CIRCUIT AND TERMINAL LAYOUT

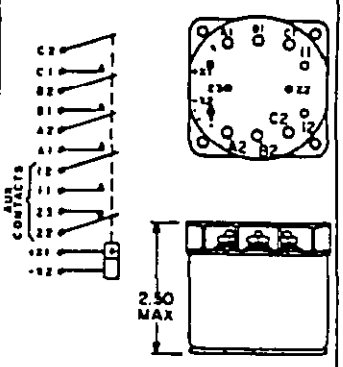
-1, -2, -11 & -12 RELAYS



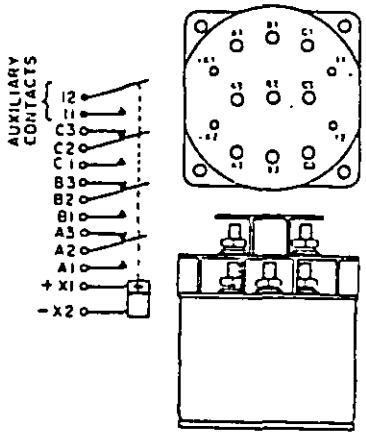
-3, -4, -13 & -14 RELAYS



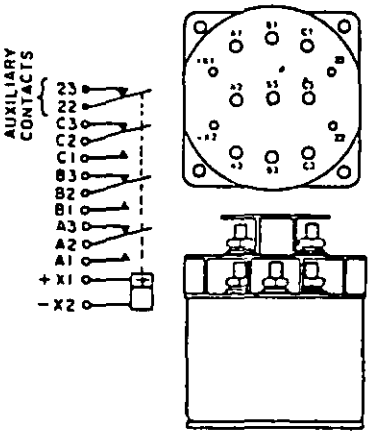
-5, -6, -15 & -16 RELAYS



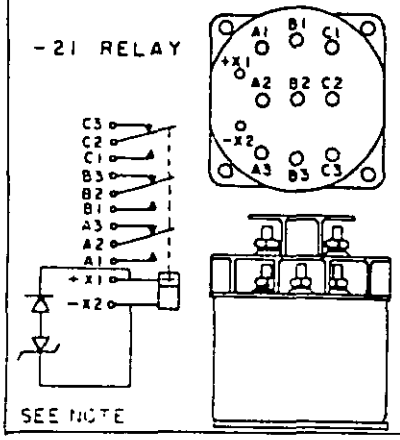
-7, -8, -17 & -18 RELAYS



-9, -10, -19 & -20 RELAYS



-21 RELAY



NOTE: When semiconductors are required, JAN TX or equivalent screened semiconductors shall be used. Relays using suppression devices shall continue to operate should the suppression circuit be in a failure mode. Diodes shall have a peak inverse voltage of 600 V dc min.

APPROVED 17 JAN 72 REVISED Ⓖ FOR CHANGES SEE PAGES 1, 4 & 6

Review activity: AF-99

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P.A AF-85 Other Cust	International interest	TITLE RELAYS, ELECTROMAGNETIC, 60 AMPERES, 3 PST OR 3 PDT, HERMETICALLY SEALED, PERMANENT MAGNET DRIVE	MILITARY STANDARD MS27751 (85)
Procurement Specification MIL-R-6106		SUPERSEDES:	PAGE 3 OF 7

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DD FORM 672 1 MAY 73

P.A. AF-85

Other Cust

Intergovernmental Interest

MIL-R-6106

SUPERSEDES:

TITLE: RELAYS, ELECTROMAGNETIC, 60 AMPERES, 3 PST OR 3 PDT, HERMETICALLY SEALED, PERMANENT MAGNET DRIVE

MILITARY STANDARD

MS27751 (85)

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FED. SUP CLASS 5945

TABLE II. Operating characteristics.

MS PART NO. MS27751 1/	COIL DATA											TIME-MILLISECONDS-MAX					
	COIL	Nominal			Max		Max. Pick-up Voltage			Drop Out Voltage 2/	Hold Voltage 2/	Operate 3/	Release 4/	Bounce			
		Volts	Freq Hz	Res Ω ±10%	Volts	Amp @ 25°C	Nominal 2/	High Temp Test	Cost Current Test					Main		Aux 5/	
														NO	NC	NO	NC
-1 -3 -5 -7 -9 -11 -13 -15 -17 -19 -21	X1X2	28	DC	200	29	.160	18	20	21	1.5	7.0	50	25	3	3	4	4
-2 -4 -6 -8 -10 -12 -14 -16 -18 -20	X1X2	115	400 6/	-	124	.100	90	95	100	5.0	35	50	80	3	3	4	4

1/ Contact transfer time @ rated voltage is 1.8 milliseconds minimum for -11 through -20.
2/ Over temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.
5/ Applicable to -5 through -10 and -15 through -20.
6/ Will operate on 115 V ac, 60 Hz with slightly higher coil current.
7/ Applicable to -21 only. Coil suppression (transient voltage back EMF) 42 V dc max.

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Procurement Specification MIL-R-6106

P.A AF-85

Other Cust

International interest

SUPERSEDES:

TITLE RELAYS, ELECTROMAGNETIC, 60 AMPERES, 3 PST OR 3 PDT, HERMETICALLY SEALED, PERMANENT MAGNET DRIVE

MILITARY STANDARD

MS27751 (85)

PAGE 5 OF 7

TABLE III. Rated contact load (amperes per pole) (case grounded).

TYPE OF LOAD	LIFE OPERATING CYCLES x 10^3	28 V dc				115 Vac, 1 Phase				115/200 Vac, 3 Phase 2/				See Appropriate Notes
		MAIN		AUX 1/		MAIN		AUX 1/		MAIN		AUX		
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	
RESISTIVE	50	50	50	3	3	60		3		60				
INDUCTIVE	20	20	20	3	3			3						
INDUCTIVE	50			2	2	60		2		60				
MOTOR	50	20	20			40				40				
LAMP	50	10	10	1	1	15		1		15				
TRANSFER, LOAD	10					12.5/60				12.5/60				3/
MECHANICAL LIFE REDUCED CURRENT	200	7	7	1	1	14		1		14				
MIN CURRENT	60	1	1			5				5				

1/ Applicable to -5 through -10 and -15 through -20.
2/ Absence of value indicates relay is not rated for 3 phase application.
3/ Transfer load indicates relay suitable for transfer between unsychronized ac power supplies of rating indicated. 12.5 amperes -1 through -10 and -21, 60 amperes -11 through -20.

APPLICATION NOTES:
1. Examination of product, external parts, shall be performed in accordance with MIL-R-6106, except the case temperature shall be limited to 150°C maximum.
2. Strength of terminals shall be performed in accordance with MIL-R-6106, except it shall be tested at room ambient temperature only.

FED. SUP CLASS 5945

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ENVIRONMENTAL CHARACTERISTICS

Temperature range	-55°C to +71°C
Max altitude rating	80,000 ft
Shock G-level	50 G's
Duration	6 ms
Max duration contact opening	10 ms
Vibration-sinusoidal	
G-level	10 G's
Frequency range	10 - 2,000 Hz
Vibration-random	
Applicable spec	NA
Power spectral density	NA
Rms G min	NA
Frequency range	NA
Curve	NA
High shock	NA
Acceleration	15 G's

ELECTRICAL CHARACTERISTICS

Insulation resistance, initial	100 megohms
After life or environmental tests	50 megohms

Dielectric strength (sea level)

	Initial	After life tests
Coil to case	1000 V rms	1000 V rms
Aux contacts	1000 V rms	1000 V rms
All other points	1500 V rms	1150 V rms

Dielectric strength (altitude)

	50,000 ft
Coil to case	500 V rms
Aux contacts	500 V dc
All other points	700 V rms

Max contact drop initial	.150 volt
After life test	.175 volt
Overload current	125 A dc, 115/200 A ac 400 Hz 400 A
Rupture current	150 A dc, 115/200 V ac, 400 Hz 500 A
Duty rating	Continuous
RFI spec	MIL-STD-461

(Applicable to coil circuits of ac operated relays)

APPROVED 17 JAN 72 REVISED (G) FOR CHANGES SEE PAGES 1,4 8 6

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TABLE IV. Qualification by similarity. 3/

MS PART NUMBER	LOADS				DYNAMICS				ENVIRONMENTAL			
	A		B		A		X		A		B	
	3PDT	3PNO	3PDT	3PNO	3PDT	3PNO	3PDT	3PNO	3PDT	3PNO	3PDT	3PNO
MS27751- 1	4				2				4			
MS27751- 2			2,3				1				2,3	
MS27751- 3		4				2				4		
MS27751- 4				2,3				1				2,3
MS27751- 5		4 1/				2				4		
MS27751- 6				2,3 1/				1				2,3
MS27751- 7	4 1/				2				4			
MS27751- 8			2,3				1				2,3	
MS27751- 9	4 1/				2				4			
MS27751- 10			2,3				1				2,3	
MS27751- 11	4 1/				2				4			
MS27751- 12			2,3				1				2,3	
MS27751- 13		4				2				4		
MS27751- 14				2,3				1				2,3
MS27751- 15		4 1/				2				4		
MS27751- 16				2,3 1/				1				2,3
MS27751- 17	4 1/				2				4			
MS27751- 18			2,3				1				2,3	
MS27751- 19	4 1/				2				4			
MS27751- 20			2,3				1				2,3	
MS27751- 21	2,3				1				2,3			

1/ Relays with auxiliary contact: Reference MIL-R-6106, Appendix, Construction Internal.
2/ Contacts and contact systems: Reference MIL-R-6106, Appendix, Contact and Contact System.
3/ Similarity exists only between dash numbers within a given column.
4/ All units must be tested: Reference MIL-R-6106, Appendix, Dynamics.

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